

MASTER YOUR CODE

State of Structure

Visualize data structures in real-time. Turn abstract logic
into concrete geometric insight.



Getting Started

Your step-by-step guide to mastering the visualizer.

Tool Core Capabilities

State Tracking

The visualizer captures every modification to your objects, allowing you to see the exact sequence of data mutation over time.

Code Mapping

Unlike standard debuggers, we focus on the structure. Watch your arrays, lists, and trees transform as your code executes.

Recording State

Initialization is the foundation. To start tracking your data, register your structure at the entry point of your logic.

```
Visualizer.register( YourDatastructure )
```

 This command hooks the visualizer into the object's memory address for continuous monitoring.

 Abstract Data Structure Visualization

Navigating Execution



Previous State

Rewind the execution to identify exactly where a logic error occurred in the data structure.



Next State

Step forward through the algorithm to observe the growth and transformation of the structure.



Inspection

Walk through code line-by-line using the dedicated app interface buttons.

Try it Yourself!

Experience the power of visualization by writing your own algorithms directly in the workspace.

Once your code is ready, simply press the **compile button** to launch the interactive viewer.

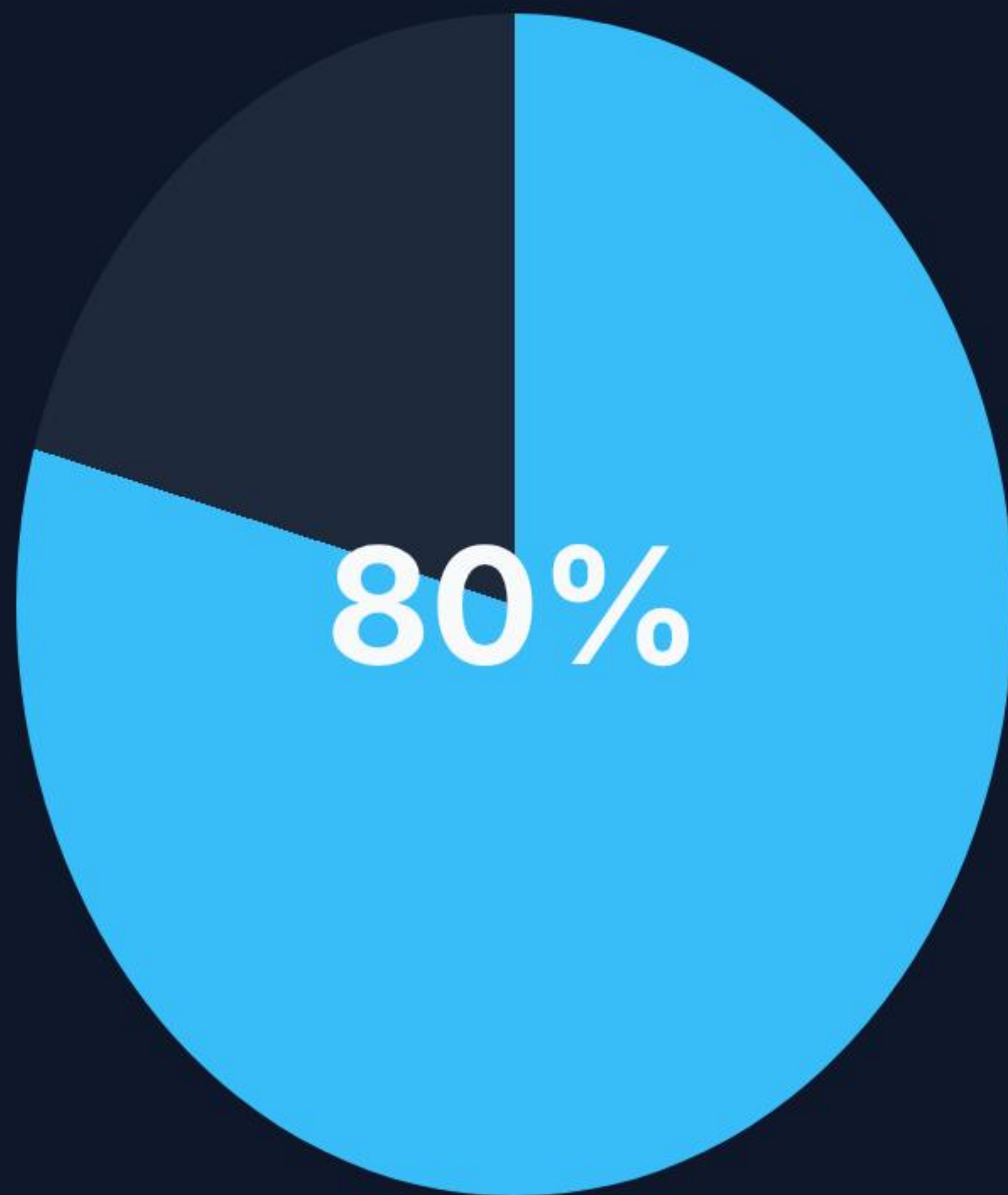
⚡ **Action:** Click Compile to Start Visualization



Compatibility Matrix

Data Structure	Visualization Mode	Recommended Use
Array / ArrayList	Linear Grid	Sorting Algorithms
LinkedList	Pointer Nodes	Dynamic Memory Ops
Binary Trees	Hierarchical Graph	Search & Traversal
Graphs	Network Topology	Pathfinding (Dijkstra)

Learning Efficiency



Visual Understanding (80%)

Theoretical Syntax (20%)

"Visual learners process data structure states 60,000x faster than reading code logs."

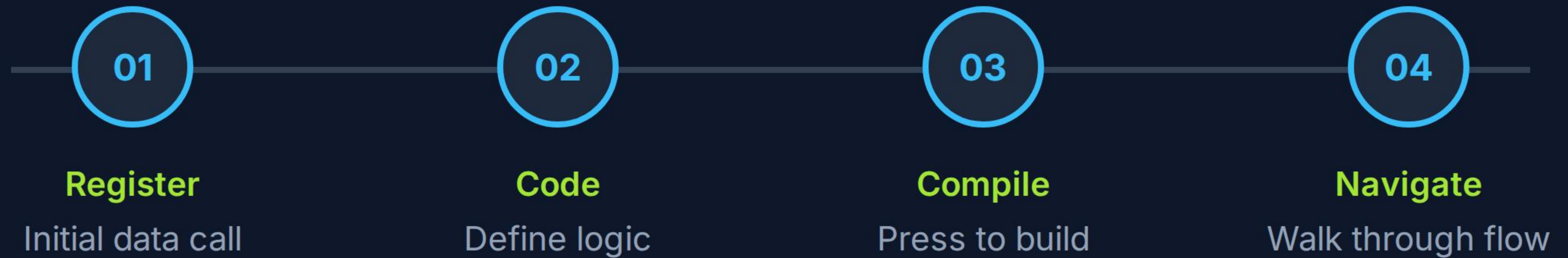
Total Object Visibility

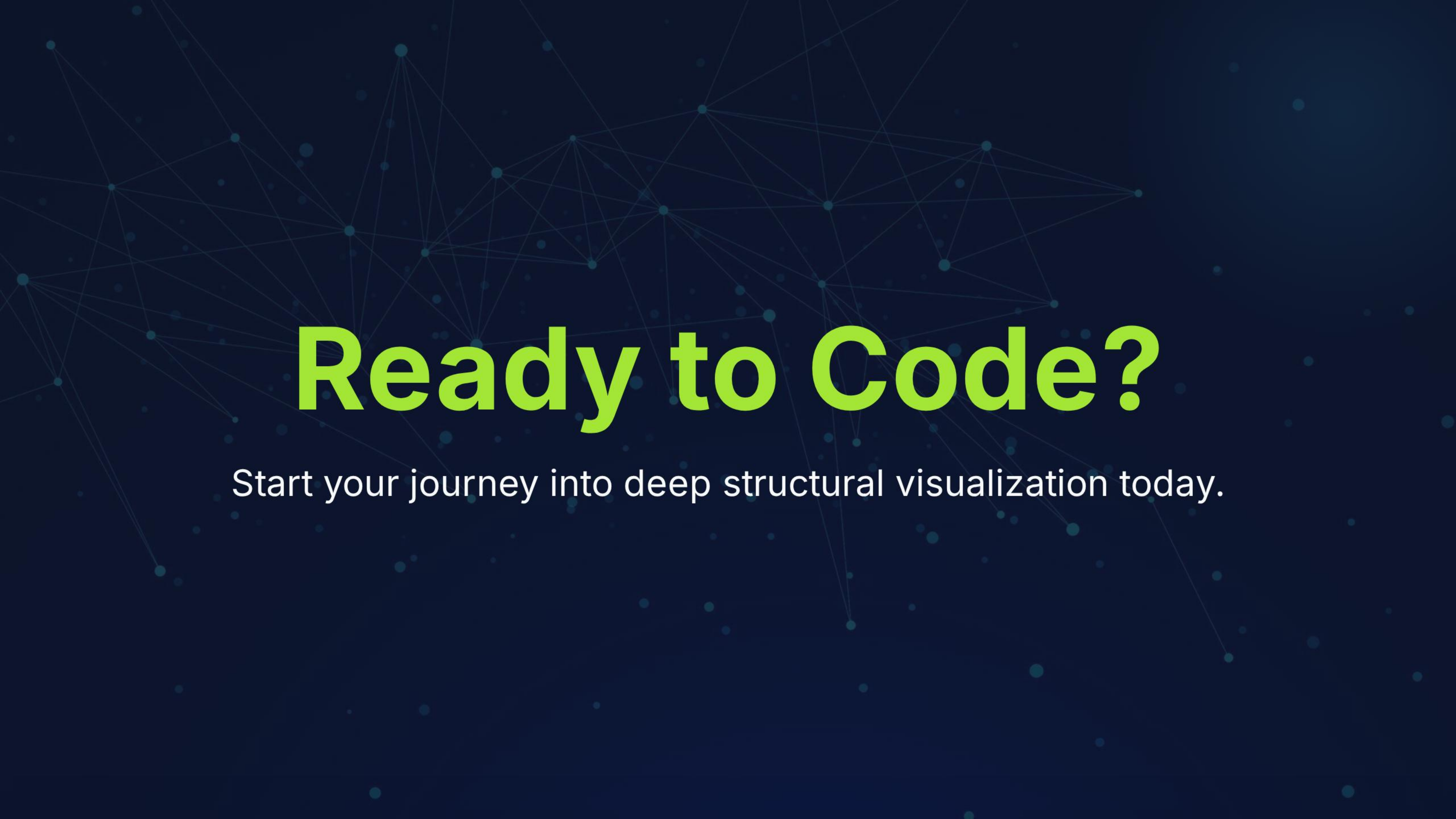
100%

State Transparency

By registering your structure, you gain complete visibility into heap allocation and variable relationships, leaving no room for "hidden" bugs.

The Master Workflow



The background features a dark blue gradient with a complex network of thin, light blue lines connecting various points. These points are represented by small, semi-transparent blue circles of varying sizes, creating a sense of depth and connectivity, reminiscent of a molecular structure or a data network.

Ready to Code?

Start your journey into deep structural visualization today.

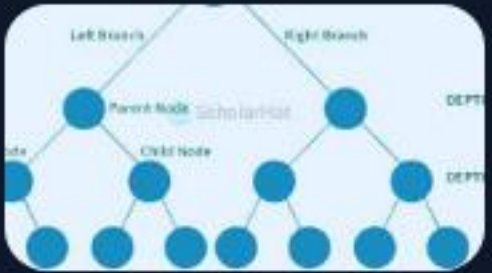
Have Questions?

Check the help documentation or reach out to the project maintainers.

Happy Coding!

State of Structure Team

Image Sources



<https://dotnettrickscloud.blob.core.windows.net/article/data%20structures/4620231214183910.png>

Source: www.scholarhat.com



<https://marketplace.atlassian.com/product-listing/files/83bf2ae-1b76-411f-a6e8-b225caa3d366?width=1840&height=900>

Source: marketplace.atlassian.com



https://static.vecteezy.com/system/resources/previews/075/392/965/non_2x/abstract-connected-glowing-nodes-and-lines-on-dark-blue-background-futuristic-network-connection-and-big-data-visualization-suitable-for-technology-innovation-and-science-vector.jpg

Source: www.vecteezy.com